

PARTS LIST

Semiconductors:

IC1, IC3	- 7805, 5V regulator
IC2	- HT12E encoder
IC4	- HT12D decoder
IC5	- UM3561 tone generator
IC6	- LM386 low-power audio amplifier
ZD1	- 3.3V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 1-mega-ohm
R2	- 51-kilo-ohm
R3	- 270-ohm
R4	- 270-kilo-ohm
R5	- 10-ohm

Capacitors:

C1, C3	- 100 μ F, 25V electrolytic
C2, C4-C6	- 100nF ceramic disk
C7	- 47nF ceramic disk
C8	- 220 μ F, 25V electrolytic
C9	- 10 μ F, 25V electrolytic

Miscellaneous:

TX1	- 433MHz transmitter module
RX1	- 433MHz receiver module
BATT.1, BATT.2	- 9V battery
S1, S2	- On/off switch
LS1	- 8-ohm, 0.5W speaker
ANT.1, ANT.2	- 17cm length single-wire antenna

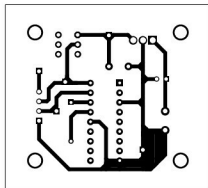


Fig. 3: Actual-size PCB of the transmitter circuit

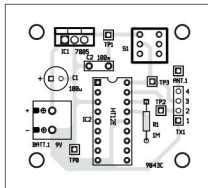


Fig. 4: Component layout of the transmitter

shown in Fig. 1. To generate clock pulses, oscillator pins 15 and 16 of IC4 are connected with an external resistance of 51k (R2).

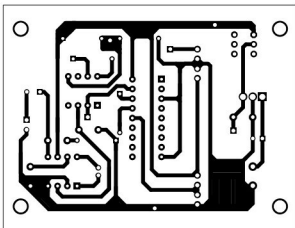


Fig. 5: Actual-size PCB of the receiver circuit

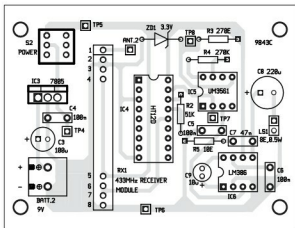


Fig. 6: Component layout of the receiver

The data output of RX1 is given to the data input DIN (pin 14) of HT12D decoder IC4. Incoming data is compared with already set address values. If valid data is received, valid transmission pin VT (pin 17) goes high and the data is accordingly transferred to the data pins. When valid data is not received, VT pin goes low, and vice-versa.

VT pin of IC4 is connected to Vss (pin 2) of UM3561 (IC5). UM3561 is a tone generator IC that gives three different tones, namely, police siren, ambulance and machine gun. These are obtained by different combinations of selection pins (SEL1 and SEL2) and oscillator pins (OSC1 and OSC2). Here, SEL1 (pin 6) and SEL2 (pin 1) of IC5 are left open to produce the police-siren tone.

Operational amplifier LM386 is used to amplify the output of IC5

to generate a louder sound. The gain of this op-amp can be from 20 to 200 by proper selection of external components between pin 1 and pin 8 of IC6. Here, the gain is set to 200. This is achieved by connecting a 10-micro farad capacitor (C9) between pin 1 and pin 8 of IC6. Output of IC5 is given to non-inverting input pin 3 of IC6 and the inverting terminal is grounded. The amplified output is taken from pin 5 through a 220-microfarad capacitor (C8) to an 8-ohm, 0.5W speaker (LS1).

During data transmission, LS1 does not produce sound because VT pin 17 of IC4 remains high to disable IC5. On the other hand, when transmission fails,

LS1 produces the sound as VT pin 17 of IC4 goes low to enable IC5.

Construction and testing

Separate PCBs are designed for transmitter and receiver units. An actual-size, single-side PCB for the transmitter circuit is shown in Fig. 3 and its component layout in Fig. 4. Similarly, an actual-size, single-side PCB for the receiver circuit is shown in Fig. 5 and its component layout in Fig. 6. Assemble both circuits separately and enclose these in two separate suitable small boxes. Ensure proper wiring to avoid any errors.

For troubleshooting, verify various voltages as given in the table. ●

All three authors are pursuing B.Tech (electronics and communication) from Maharaj Vijayaram Gajapathi Raj College of Engineering, Vizianagaram, Andhra Pradesh